



Land

Early steam



Stack releases exhaust smoke

Pen-y-darren locomotive UK 1804

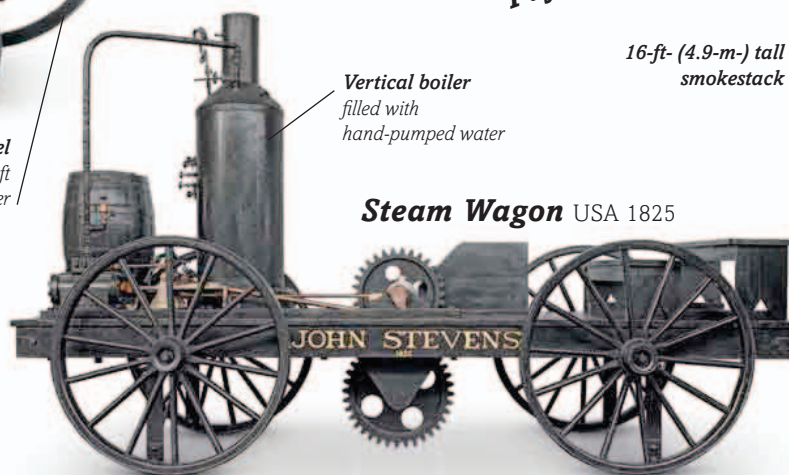
Giant flywheel measures 8 ft (2.45 m) in diameter



Towing hook

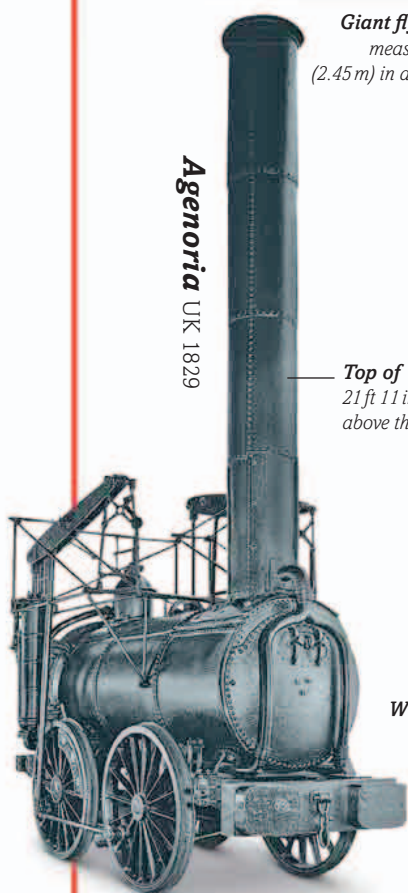
Puffing Billy UK 1813

16-ft- (4.9-m-) tall smokestack



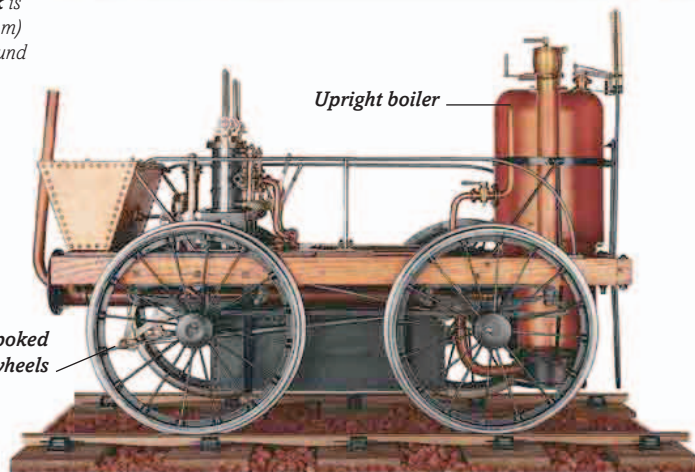
Steam Wagon USA 1825

JOHN STEVENS



Agenoria UK 1829

Top of stack is 21 ft 11 in (6.7 m) above the ground



Upright boiler

Wire-spoked wheels

Novelty UK 1829

Rocket UK 1829



The first steam engines were used in factories to run machines, or in mines to pump out water. Richard Trevithick, a mining engineer, was one of the first to use steam to power a moving locomotive, sparking a transportation revolution.

In Wales in 1804, Trevithick's *Pen-y-darren* made the first railroad journey at less than 2 mph (4 km/h), hauling 12 tons of cargo and 70 people over 8.9 miles (14.4 km). Other steam engines, such as the *Puffing Billy* and *Agenoria*, quickly followed, ferrying coal or goods from factories.